

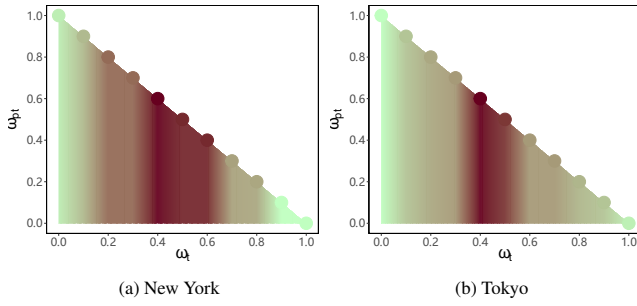


Figure 9: The analysis of reward of the temporal agent.

The Study of Mobility Hypergraph (Q4)

The design of Mobility Hypergraph. We conduct an ablation experiment to verify the necessity of the hypergraph architecture. In this experiment, we design a variant of STI-HRL called STI-HRL', which directly takes the user's sequence as input without using a hypergraph architecture. As shown in Figure 6, STI-HRL shows a clear performance advantage compared to STI-HRL'. This result highlights the effectiveness of the hypergraph architecture in dealing with such problems.

The effects of hyperedge types. We take the embeddings of three types of hyperedges (POI, Time, and Zone) as the state. Therefore, we introduce six variants of STI-HRL to study the effects of hyperedge types, including (1) STI-HRL- P' , POI hyperedges only; (2) STI-HRL- T' , time hyperedges only; (3) STI-HRL- Z' , zone hyperedges only; (4) STI-HRL- ZT' , zone and time hyperedges; (5) STI-HRL- TP' , time and POI hyperedges; (6) STI-HRL- ZP' , zone and POI hyperedges. We conduct the ablation study on the six variants and present the results in Figure 7. The results demonstrate that varying permutations and combinations of hyperedge types yield different performances. This indicates the significance of the semantics conveyed by these hyperedge types in representing human mobility events. Only STI-HRL, which incorporates all hyperedge types, achieves optimal performance.

The Analysis of Reward Design (Q5)

We consider different combinations of weight setting of reward functions to test the performance of STI-HRL. Since the high-level reward settings (w_S and w_T) are automatically learned, which cannot be manually adjusted. Therefore, we only analyze the reward setting of spatial and temporal agents. Specifically, for the spatial agent reward r_S , three components (r_d , r_c , r_p) are considered. We project the performance of different combinations of w_d , w_c , and w_{ps} ($w_d + w_c + w_{ps} = 1$) to 3D space, and mark the performance in different scales of color, as shown in Figure 8. The better the performance, the darker the color. For the temporal agent reward r_T , two components (r_t and r_p) are considered. We plot the results in Figure 9.

Related Work

Human Mobility Prediction

Human mobility modeling centers on predicting users' next-visit locations by harnessing insights from historical check-in data. The rise of deep learning introduced advanced techniques like Recurrent Neural Networks (RNNs) (??) and self-attention mechanisms (??). While they excel at modeling sequential regularities, especially with spatial-temporal data, they tend to oversimplify complex spatial-temporal interplay influencing user choices.

Hypergraph Learning

Compared to other graph models (???) that deal with heterogeneous information, hypergraph learning (???) offers a robust representation of the nuanced information within human mobility data, outshining traditional graph-based counterparts. Yet, even with advances in hypergraph embedding, many methods overlook daily behavioral similarities among unconnected individuals as seen in historical trajectories (?), curbing the full potential of existing models.

Reinforcement Learning

At its core, reinforcement learning focuses on learning policy learning through environmental interactions and feedback (??). The duality of exploration and exploitation in this learning paradigm aptly captures the evolving user preferences. Hierarchical Reinforcement Learning (HRL) extends it to a wide variety of comprehensive recommendation domains (?????) has a wide range of applications. Our study pioneers the application of HRL for the next visit in human mobility, approaching it from a unique disentangled viewpoint. Existing HRL models aren't directly transferrable to our challenge due to distinct goals and data structures.

Conclusion

In this study, we address the challenge of capturing the spatial-temporal interplay in human mobility. We frame this challenge as a two-tiered decision-making task: the low-level focuses on disentangling spatial and temporal preferences, while the high-level synthesizes these insights to produce a comprehensive mobility decision. Our approach leverages a hierarchical reinforcement learning framework. To effectively encapsulate the multifaceted semantics of spatial-temporal dynamics, we introduce a Mobility Hypergraph to structure the mobility records, utilizing hyperedge embeddings as states for policy learning. Comprehensive experiments validate the efficacy of our method. Its consistent superiority over baseline models underscores the importance of spatial-temporal interplay in human mobility modeling.

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